

LESSON 130 (0.5) Ground (2.0) Local Dual (1.0) Instrument

Lesson Objective

The purpose of this lesson is to introduce maneuvering solely by reference to flight instruments and review basic radio navigation techniques and recoveries from unusual attitudes to facilitate aeronautical decision making and situational awareness in an airplane. In addition, the applicant will practice short and soft-field takeoff and landings, emergency descents, and steep turns.

Pre-Brief

- Review Maneuvers and Procedures to be Performed in Flight
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- During this lesson quiz the student on the following and provide instruction where needed.
- Student questions.

References

- Airplane Flying handbook (FAA-H-8083-3C)
 - Chapter 6: Takeoffs and Departure Climbs
 - Chapter 9: Approaches and Landings
 - Chapter 10: Performance Maneuvers
 - Chapter 18: Emergency Procedures
- Instrument Flying handbook (FAA-H-8083-15B)
 - Chapter 7: Basic Flight Maneuvers using analog/Electronic Flight Display

Instructor Guidance

- Verify the student has a view-limiting device before departing. If they arrive without one, the instrument portion of the flight cannot be conducted — address this during the briefing before heading to the aircraft.
- Before the flight, confirm the student has a solid understanding of both performance landing procedures and the instrument cross-check scan. Quiz them briefly on the radial scan technique and the short-field and soft-field completion standards. A student who is shaky on either area will have difficulty managing the workload of this lesson, which combines multiple demanding skill sets in a single flight.
- Discuss motion sickness proactively before the flight, particularly if conditions are turbulent
- For VOR navigation in the actual aircraft, take the time to walk through the tuning and identification process step by step before beginning the intercept. Students who are unsure of which button switches between VLOC and GPS on the G5 or Avidyne will lose situational awareness quickly while trying to navigate under the hood.

Lesson Content

Checklist Procedures

- Have the student run through all checklists.
- Incorporate flows for memory items.
- Student should be able to perform all briefings.
 - Setting correct flap configuration for line item in Before Takeoff Check
 - During Takeoff Briefing have student brief the type of takeoff being done

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units

- Ensure that the student understands how to properly use COMS and NAV units.
- Introduce how to put in NAV frequencies in the G430 or G650
- Show student how to correctly tune and ID and VOR using the corresponding NAV button to listen and the volume button to ID.
 - When operating with a G650, show student how it automatically identifies a station by showing the corresponding identifier next to the NAV frequency.

Engine Starting

- Correctly determine and use appropriate start procedure.
- Incorporate the use of flows.

Radio Communication Procedures and ATC Phraseology

- The student should be able to conduct radio calls in towered airport.
- The student should be able to conduct radio calls in the practice area.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Taxiing

- The student should be able to taxi on the centerline at appropriate speeds on the ramp and taxiways without instructor guidance
- Ensure the correct use of wind correction throughout taxi.

Short-Field Takeoff and Maximum Performance Climb

- Student does, student says:
 - Verify correct procedure.
 - Ensure that the student holds the brakes until full static RPM.
 - Verify that the climb out is conducted at 63 knots.
 - Verify that the student says obstacle cleared if there is an obstacle.

- Verify that the student says positive rate of climb before retracting the flaps.

Soft-Field Takeoff and Climb

- Student does, student says:
 - Verify correct procedure.
 - Coach the student through maintaining full aft stabilator and staying in ground effect.
 - Verify that the climb out is conducted at 63 or 79 knots depending on obstacles.
 - Verify that the student says obstacle cleared if there is an obstacle.
 - Verify that the student says positive rate of climb before retracting the flaps.

Basic Instrument Maneuvers

- Refer to the fundamental skills of Basic Attitude Instrument flight, as well as control and performance methods sections in lesson 127.

Straight-and-Level Flight (Instrument)

- Establish and maintain constant heading and altitude using attitude + power = performance.
 - Around 2.5 degrees of pitch + around 2300 RPM = Straight and Level at 90kts
- Pitch control:
 - Adjust pitch to maintain constant altitude (small, smooth yoke movements).
 - Lower pitch attitude + higher RPM = higher airspeed
 - Higher pitch attitude + lower RPM = lower airspeed
- Bank control:
 - Use the heading indicator to maintain heading.
 - Cross-check the turn coordinator or ball to ensure coordination and detect small banks early.
 - Make small aileron and rudder corrections together to maintain wings level.
- Power control:
 - Set and hold the power setting appropriate for the desired airspeed (e.g., 2300–2400 RPM in a Warrior).
 - Use airspeed indicator as the primary power instrument in level flight.
- Trim:
 - Trim off all control pressures once altitude and airspeed stabilize.
 - Proper trim reduces workload and helps maintain precise control.
- Instrument Cross-Check and Interpretation:
 - Begin with a selective radial scan — attitude indicator as hub; supporting instruments as spokes.
 - Correct for small deviations promptly and smoothly (½-bar width corrections on the attitude indicator).
 - Verify performance by comparing multiple instruments (altimeter, VSI, airspeed).
- Common tendencies:
 - Over-controlling due to fixation on one instrument.
 - Failure to recognize small bank or pitch changes early.

- Slow power response leading to airspeed/altitude drift.
- Emphasize the need to scan!

• Constant Airspeed Climbs (Instrument)

- Use attitude + power = performance; pitch controls airspeed, and power controls rate.
- Select a specific pitch attitude on the attitude indicator that yields the desired airspeed
 - V_Y : About a 10 degrees pitch attitude + full power = climb at 79 knots.
 - V_{CLIMB} : About a 5 - 7.5 degrees of pitch attitude + full power = climb at 87 knots.
- Apply smooth power to full while maintaining coordinated flight with rudder.
- Cross-check primarily with the attitude indicator, and reference supporting instruments:
 - Airspeed indicator – confirms constant climb speed.
 - VSI and altimeter – confirm climb performance trend.
 - Heading indicator + turn coordinator – ensure wings level.
- Trim off control pressures once the airspeeds stabilizes.
- Anticipate level-off approximately 10% of the rate of climb (for example, 50 feet per 500 fpm).
- Transition by reducing pitch and power smoothly to cruise settings while maintaining constant airspeed.

• Constant Airspeed Descents (Instrument)

- Use attitude + power = performance; pitch controls airspeed, and power controls rate.
- Select a specific pitch attitude on the attitude indicator that yields the desired airspeed, adjusting power and pitch for desired airspeed
 - Nose right under the horizon line (attitude indicator) + power idle = glide descent at 73kts
 - Nose at -2.5 degrees pitch (attitude indicator) + 1600-1800 RPM = descent at 90kts
- Adjust power to desired RPM while maintaining coordinated flight with rudder.
- Cross-check primarily with the attitude indicator, and reference supporting instruments:
 - Airspeed indicator – confirms constant descent.
 - VSI and altimeter – confirm descent performance trend.
 - Heading indicator + turn coordinator – ensure wings level.
- Trim off control pressures once the airspeed stabilizes.
- Anticipate level-off approximately 10% of the rate of climb (e.g., 50 feet per -500 fpm).
- Transition to level flight by increasing pitch and power smoothly to cruise settings while maintaining constant airspeed.

• Turns to Headings (Instrument)

- Establish appropriate bank angle for a standard rate turn
- To calculate how many degrees of bank a standard rate of turn is, use the following formula:
 - Divide airspeed by 10 and add $\frac{1}{2}$ of the result.
 - Warrior standard rate:

- 90/10 = 9.
 - $9/2 = 4.5$.
 - $4.5 + 9 = 13.5$ degrees = standard rate.
 - Use the attitude indicator to set and hold bank angle; turn coordinator confirms standard-rate turn (3 degrees per second)
 - A 180 degree turn takes 1 minutes and 360 degree turn takes 2 minutes
 - Primary instruments:
 - Heading indicator – primary for bank when stabilized.
 - Attitude indicator – establishes bank attitude.
 - Altimeter + VSI – monitor altitude; adjust pitch to maintain level flight.
 - Lead the rollout by $\frac{1}{2}$ the bank angle in degrees (e.g., 10° early from a 20° bank).
 - Maintain constant power and coordinated rudder throughout turn
 - Remind the student that as the bank increases, pitch must increase to compensate for the reduction in vertical lift.
 - Cross-check scan: Attitude → Heading → Altimeter → Turn Coordinator → Repeat.
- **Recovery from Unusual Flight Attitudes (Instrument)**
 - Recognize an unusual attitude by instrument interpretation (not “feel”):
 - Nose-high: Increasing pitch, decreasing airspeed, increasing altitude.
 - Nose-low: Decreasing pitch, increasing airspeed, altitude loss, possible turn indication.
 - Recovery (Nose-High):
 - Apply maximum power.
 - Lower the nose to prevent stall — relax back pressure smoothly.
 - Level the wings with coordinated aileron and rudder.
 - Return to desired attitude and trim.
 - Recovery (Nose-Low):
 - Reduce power to prevent overspeed.
 - Level the wings using coordinated aileron and rudder.
 - Gently raise the nose to the level flight attitude.
 - Re-trim and resume normal scan.
 - EMPHASIZE wings level before pitch up; cover what happens when load factor increases
 - Emphasize instrument cross-check (attitude → airspeed → altimeter → VSI → heading → turn coordinator).
 - Avoid over-controlling; make small, smooth corrections.
 - Reinforce the “power + attitude + trim” relationship throughout recovery.
 - **Radio Communications, Navigation Systems/Facilities, and Radar Services (Instrument)**
 - Radio Communication
 - Use standard phraseology and proper radio technique (listen before transmitting, think before keying).
 - Ensure clear, concise, and professional transmissions

- include who you're calling, who you are, where you are, and what you want.
- Acknowledge and read back critical instructions (altitude, heading, altimeter settings, runway assignments).
- Maintain situational awareness by monitoring guard (121.5) or ATIS/CTAF as appropriate.
- Always verify frequency changes; anticipate upcoming ATC handoffs or frequency zones.
- Discuss radio calls between the pilot and ATC when inadvertently entering IMC conditions.
- Navigation Systems/Facilities
 - Discuss how GPS can be used after encountering IMC conditions
 - Using nearest page, NAV, and AUX page
 - Demonstrate various ways of proceeding direct to an airport
 - Direct button → input the airport identifier → enter → enter.
 - Nearest page → nearest airports → direct → enter → enter.
 - For the G5:
 - Press the CDI button on the G430 to switch it from VLOC to GPS.
 - The G5 HSI should then show the magenta GPS needle.
 - For the Avidyne:
 - Press the up-most left bezel key on the PFD. This will cycle between GPS1, VLOC1, GPS2, and VLOC2. Select the desired GPS.
 - For the purpose of this lesson, the student should be able to:
 - Proceed direct to an airport.
 - Intercept and track a GPS course.
 - Discuss how instrument approaches work (surface level)
- Demonstrate how to use the VOR.
 - Set the VOR frequency in the G430 or G650 (show where to find the frequency).
 - Tune and ID the VOR by pressing the ID button and NAV1 or NAV2 (depending on which NAV is being used).
 - Verify with the morse code on a sectional chart.
 - For the G5:
 - Once tune and ID is complete, press the CDI button on the G430 or G650 to switch it from GPS to VLOC.
 - The G5 HSI should then show the green VOR needle.
 - To adjust the course, press the right knob on G5, scroll to the right to CRS, and press the right knob. This allows the pilot to twist to the desired course.
 - For the Avidyne:
 - Once tune and ID is complete, press the up-most left bezel key on the PFD. This will cycle between GPS1, VLOC1, GPS2, and VLOC2. Select the desired VLOC.
 - To adjust the course, use the left knob on the PFD to adjust the needle to the desired course. To center it, simply press the left knob.

- Cover the difference between an outbound and inbound indication.
 - Outbound: arrows point away from each other.
 - Inbound: arrows point the same way.
 - When given an outbound instruction like “TCF609 intercept and track the 360-radial outbound”, the course should be set to 360.
 - When given an inbound instruction like “TCF609 intercept and track the 360-radial inbound”, the tail should be set to 360 OR take the reciprocal of 360 and set that number to the course.
- For the purpose of this lesson, the student should be able to:
 - Figure out the current radial.
 - Intercept and track inbound/outbound instructions using an intercept angle of 30-45 degrees in the correct direction.
- Radar Services
 - Understand primary radar (range and azimuth only) and secondary radar (transponder-based — Mode A/C/S).
 - Transponder codes:
 - 1200 – VFR, 7500 – Hijack, 7600 – Radio Failure, 7700 – Emergency.
 - When working with ATC radar:
 - Listen for assigned discrete transponder code.
 - Respond to radar vectors using standard-rate turns and precise altitude control.
 - Read back headings and altitudes to ensure clarity.
- **Instrument errors**
 - Cover how to recognize an instrument error/failure and how to solve it
 - Pitot-static blockages:

Situation / Instrument	Airspeed	Altimeter	VSI
Blocked pitot	Zero	Works	Works
Blocked pitot and drain hole	High in climb Low in descent	Works	Works
Blocked static	Low in climb High in descent	Frozen	Frozen
Alternate static	Reads high	Reads high	Momentary climb

- ADAHRS failure:
 - Red X's.
 - Reset the circuit breaker (has to be done within 30 seconds of the AHRS failure).
 - If PFD function does not return then use the standby instruments, contact ATC, and land as soon as practical.
- DAU failure (Avidyne):
 - Red X's.

- Reset the circuit breaker.
- If MFD function does not return then contact ATC, and land as soon as practical.

Steep Turns

- Procedure: Review Maneuvers Manual
- Instructor does, student says:
 - Entering the turn:
 - Ensure pre-maneuver check is complete.
 - Choose a visual reference point and bug it in the HSI.
 - Smooth level turn to 45 degrees of bank (too aggressive with the bank and back pressure can result in an accelerated stall!)
 - At 30 degrees of bank, add 100-200 RPM and 3 rolls of aft trim.
 - Maintaining the turn:
 - Focus should primarily be outside.
 - Show the student where the horizon intersects with the panel/cowling. The student should maintain that sight picture.
 - Coordination.
 - Reference the instruments.
 - Corrections:
 - If the nose falls below the desired sight picture, pitch needs to increase.
 - If the nose rises above the desired sight picture, pitch needs to decrease.
 - Power should only be used to correct for airspeed changes.
 - Rolling out:
 - Roll-out should commence at $\frac{1}{2}$ the bank angle.
 - Pitch the nose down to counteract the decrease in horizontal lift and increase in vertical lift.
 - Bring the power back to cruise power.
 - Trim for level flight.
 - Situational Awareness
 - Look outside primarily at the horizon reference point on the cowling relative to the horizon.
 - Divide attention between outside references and instrument cross-check (altimeter, VSI, airspeed).
 - Smooth, continuous control inputs show mastery of aircraft handling under varying load conditions.
- Student says, Student does:
 - Verify Correct Procedure
 - Conduct steep turns with instructor guidance and maintain altitude ± 150 ft, airspeed ± 15 knots, bank angle $\pm 10^\circ$ and roll out on the entry heading ± 15 degrees.

Emergency Descent

- Procedure

- Review Maneuvers Manual
- Talk about the following fundamental concepts
 - Aerodynamics of descent
 - An emergency descent involves placing the airplane in a steep, controlled descent at a safe airspeed.
 - A bank of approximately 30 to 45 degrees helps increase drag and prevents excessive airspeed buildup.
 - The nose-down attitude decreases lift and increases the rate of descent. The pilot must balance pitch and bank to achieve a rapid descent without exceeding structural limits.
- Load Factor
 - As the aircraft banks and descends, a positive load factor is produced due to increased back pressure required to maintain a controlled path. Typical load factors range from 1.2 to 2.0 G.
 - Maintaining a positive load factor ensures the airplane remains in a stable, loaded condition rather than free fall.
 - Excessive back pressure should be avoided to prevent exceeding load limits or inducing an accelerated stall.
- Airspeed Control
 - The pilot must maintain airspeed at 95 kts. Airspeed should be monitored continuously to prevent flap overspeed (V_{fe}).
- Student says, student does:
 - Verify correct procedure.
 - Conduct emergency descents while maintaining airspeed ± 10 knots, using a bank angle between 30° and 45° , and levelling off at a specified altitude ± 100 ft.

Go-Arounds or Rejected Landings

- Have your student practice go-around procedures.
- Encourage your student to call for a go-around themselves rather than waiting for you to call it.
 - If you do call for a go-around, the student should immediately execute the procedure rather than discussing it first.

Short-Field Approach and Landing

- Student does, student says:
 - Verify correct procedure.
 - Coach the student through staying higher on the approach and slower.
 - Coach the student through maintaining a steeper rate of descent and focusing on the aiming point.
 - Help on the round out and flare if needed.
 - Emphasize the importance of not forcing the airplane on the ground. If the point cannot be made, perform a go-around.

- Verify "maximum" braking.

Soft-Field Approach and Landing

- Student does, student says:
 - Verify correct procedure.
 - Coach the student through the round out, staying in ground effect, and flaring.
 - Verify minimal braking.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Completion Standards

This lesson is complete when the applicant can:

1. With minimal instructor guidance:
 - a) Use radio aids for orientation.
 - b) Track and bracket radials.
 - c) Identify and practice proper corrective actions to unusual attitudes.
 - d) Maintain altitude ± 150 ft, heading ± 20 degrees, and airspeed ± 10 knots.
 - e) Conduct short field landings and touch down at or within 400 ft beyond a specified point.
 - f) Conduct soft field landings and touch down softly with no drift.
 - g) Conduct emergency descents while maintaining airspeed ± 10 knots, using a bank angle between 30° and 45° , and levelling off at a specified altitude ± 100 ft.
 - h) Conduct steep turns and maintain altitude ± 150 ft, airspeed ± 15 knots, bank angle $\pm 10^\circ$ and roll out on the entry heading ± 15 degrees.
2. Without instructor guidance:
 - a. Maintain directional control during takeoffs and landings.
 - b. Maintain a stabilized approach.